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Erwin et al.

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[54] **METHOD OF MAKING A FIBER REINFORCED RACKET**

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[52] **U.S. Cl.**..... **156/185,** 156/189, 156/190, 156/196, 156/245, 156/285, 264/313, 264/314, 273/73 F

[51] **Int. Cl.** **B65h 81/00,** A63b 49/00

[58] **Field of Search**..... 156/162, 173, 175, 156/184, 189–190, 196, 285, 156, 191, 245, 287, 185; 264/314, 313; 273/73 F

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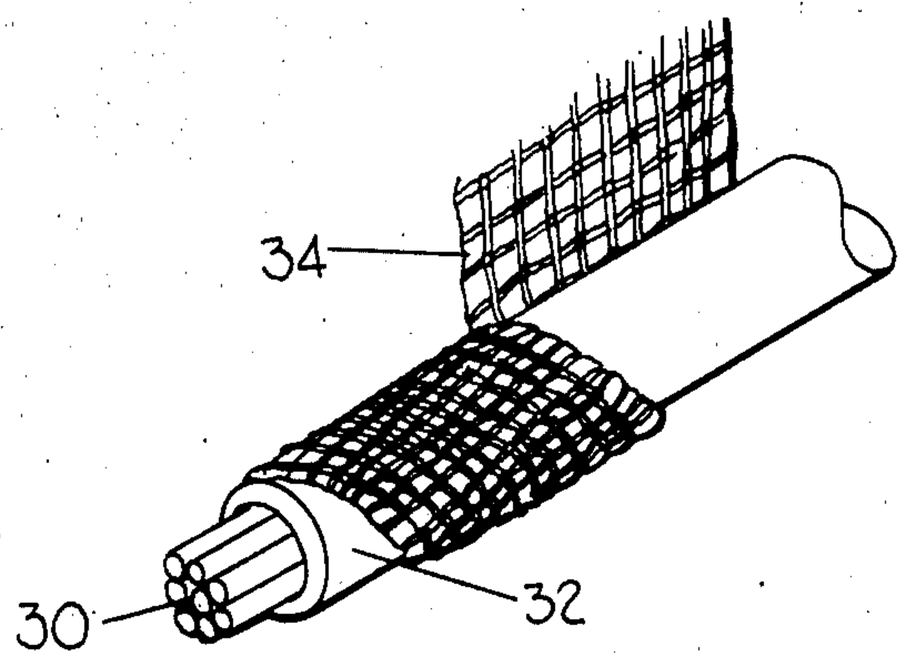
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[57] **ABSTRACT**

Racket having frame including an oval head portion and a handle portion integrally formed by a tubular member composed of helically wound fibers of high tensile strength embedded in a hardened binder and having a pre-formed reinforcing member defining the base of the oval head portion and bonded on opposite sides to the tubular member, the handle portion being defined in part by generally parallel extending portions of the tubular member surrounded by a grip.

Method of making a tubular article by helically winding high tensile strength fibers about a core provided with a substantially fluid tight casing; the core is removed and the wound casing is placed in a mold, a binder for the fibers is added, the interior of the casing is supplied with fluid under pressure, and the binder is hardened.

4 Claims, 8 Drawing Figures



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2 Sheets-Sheet 1

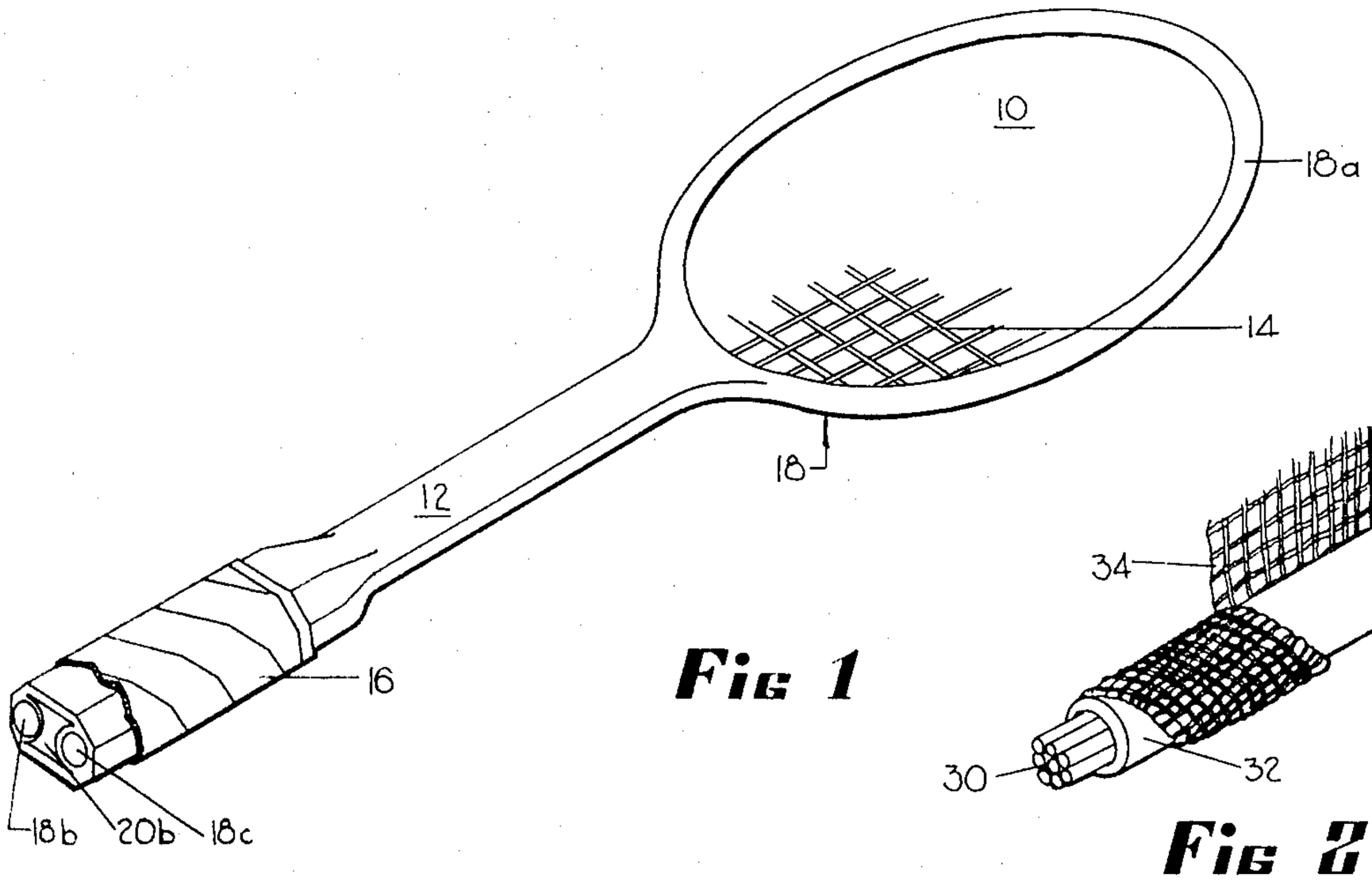


Fig 1

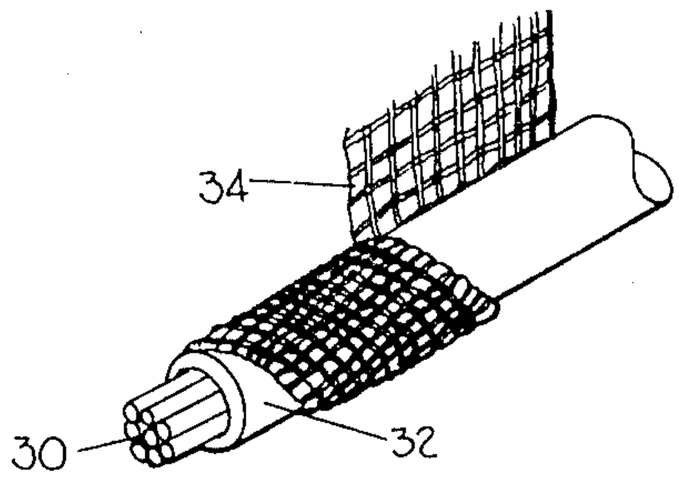


Fig 2

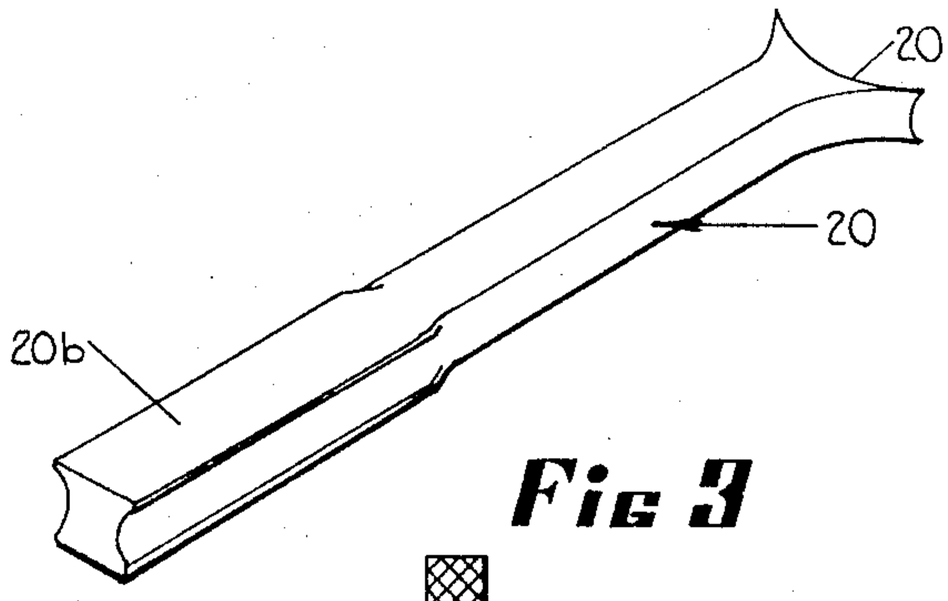


Fig 3

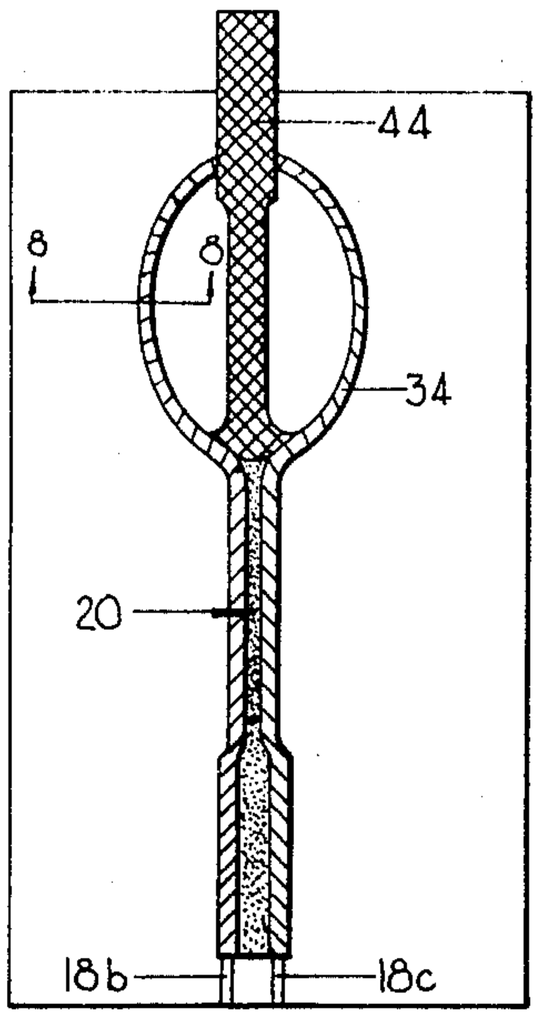


Fig 5

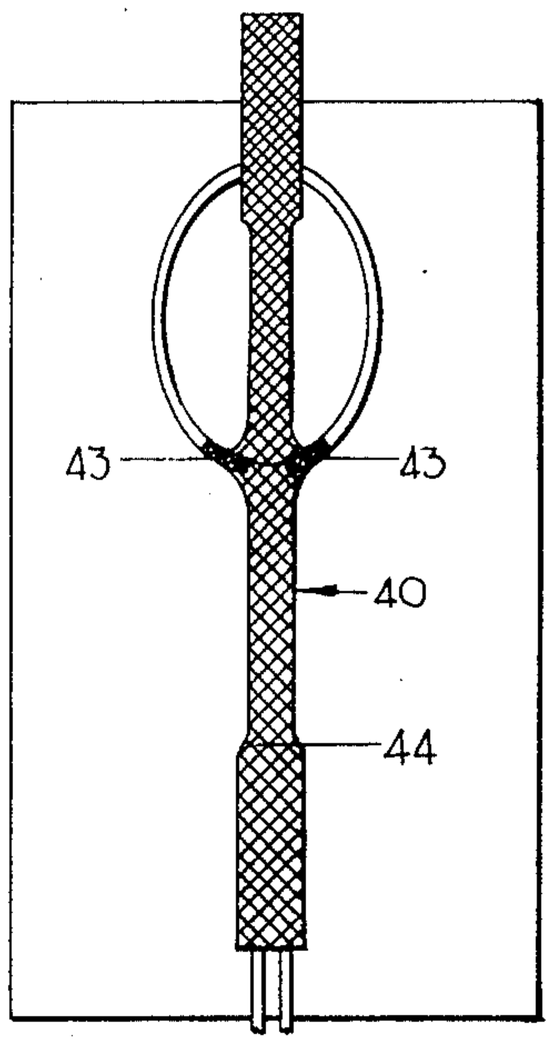


Fig 4

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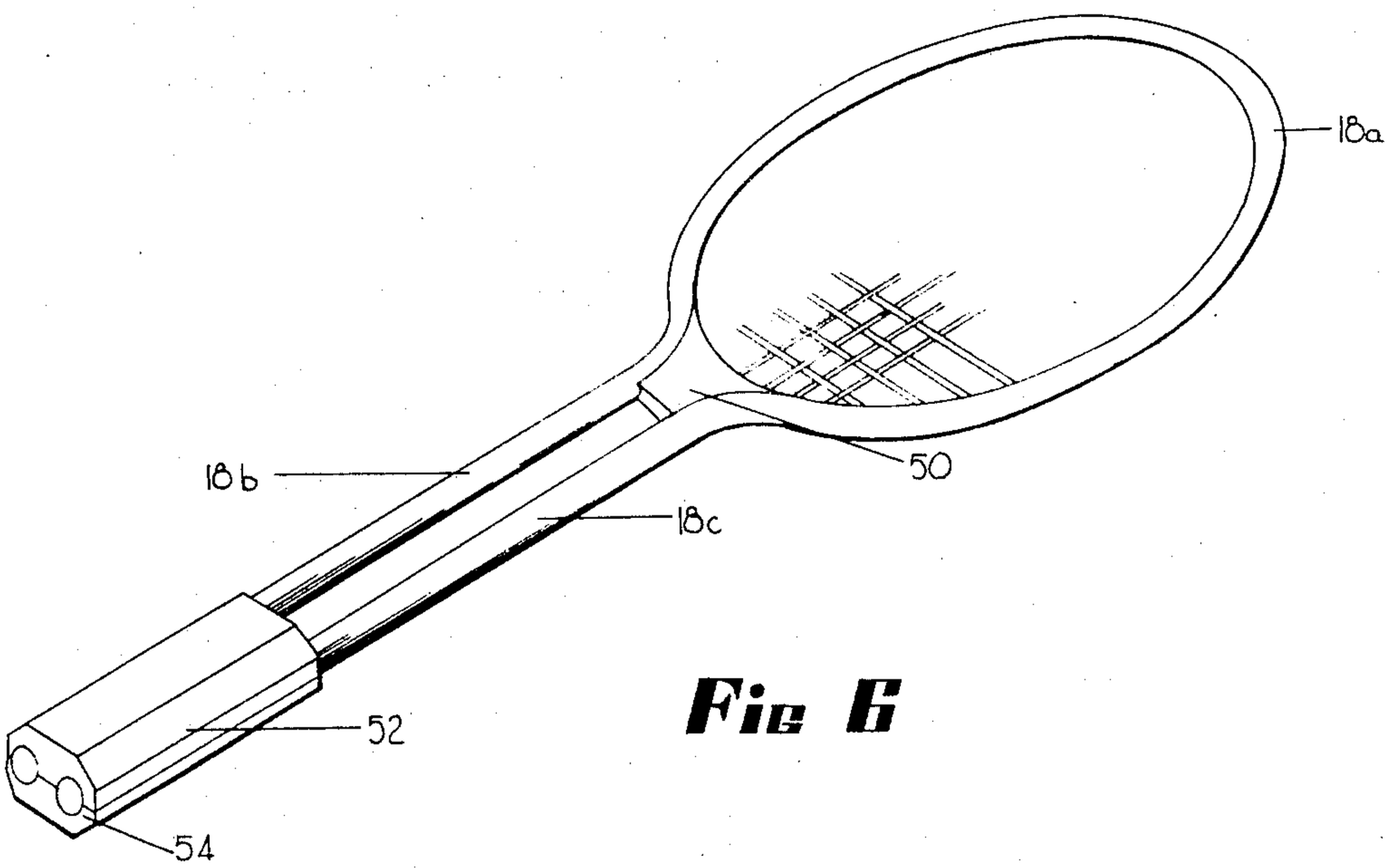


Fig 6

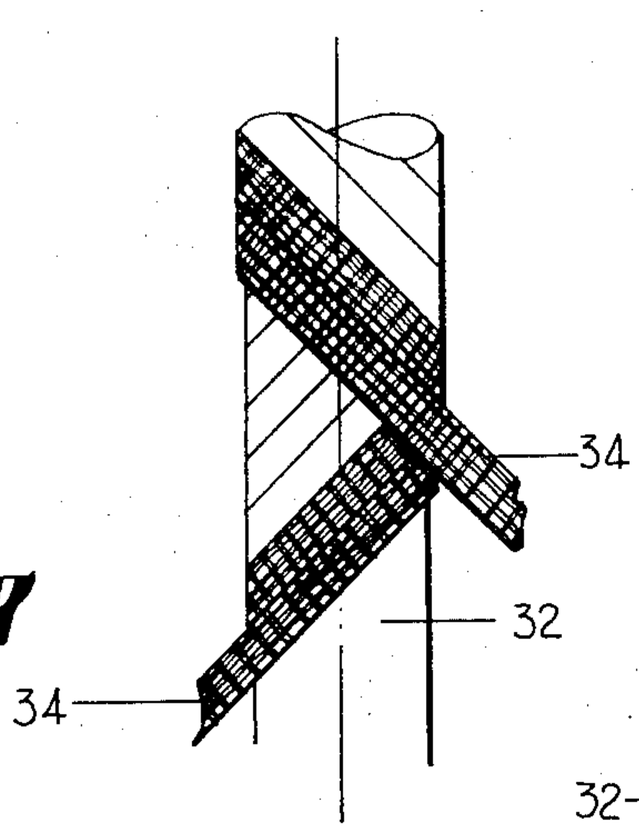


Fig 7

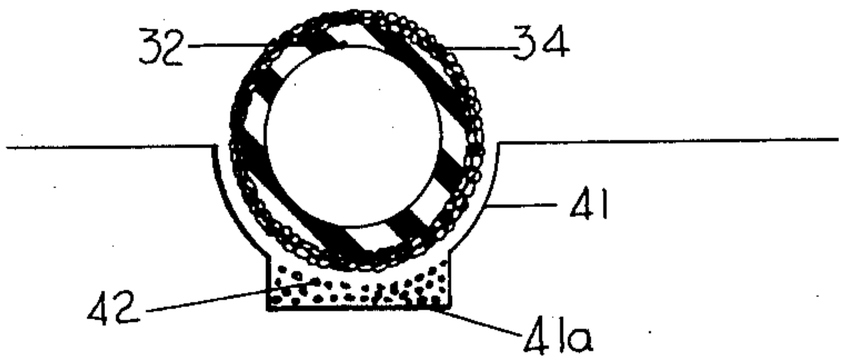


Fig 8

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METHOD OF MAKING A FIBER REINFORCED RACKET

BACKGROUND OF THE INVENTION

This invention relates to a racket for various games such as tennis, squash, and the like. While the description will primarily be in terms of a tennis racket, it should be understood that the principles taught are applicable to any type of strong racket, and to the production of other curved, tubular articles.

Until relatively recently, all good rackets were made substantially of wood, generally in the form of a plurality of curved pieces laminated together by glue or the like. These rackets possessed many desirable qualities from the standpoint of strength, lightweight, etc. However, by the same token, these rackets were subject to certain disadvantages.

In the first place, wood is always subject to warping, particularly under the stress of tightly stretched strings.

In addition, and more importantly, the manufacturing process was by necessity largely a hand operation. It therefore follows that this construction is both expensive and time consuming. Comparatively recently, various workers in the art have produced rackets wherein the frame is constructed of steel or aluminum. Obviously, a racket frame of such a material does not warp, and possesses a very high degree of strength. Unfortunately, none of the steel and aluminum rackets developed to date can be strung originally, or perhaps more importantly, repaired in the conventional manner wherein a pick is inserted in a hole in the racket frame to hold the string while it is being threaded through the next hold. Thus, the stringing of a steel or aluminum racket is difficult and very time consuming. Furthermore, the initial cost of materials, as well as the cost of the equipment for forming the metal racket frames is high, thereby making the consumer cost of such a racket very high.

The use of glass reinforced plastic materials has become widespread during this same period of time. It is well known that the glass reinforced plastics have a very high strength, they have a good modulus of elasticity, the raw materials are inexpensive, and it can be readily formed and otherwise handled.

Yet in spite of this knowledge, the art has been unable to develop a satisfactory racket frame formed of a fiber reinforced plastic material.

SUMMARY OF THE INVENTION

The method of this invention, generally considered, includes the following steps. First of all, a fluid tight casing is provided with a core. The core and casing must have sufficient rigidity that they can be covered by a helically wound fiber or fibers having the necessary tensile strength. This core will be removed at any convenient stage in the process, leaving the fluid tight casing in place. The wound casing and a suitable binder are then placed in a mold having a cavity configured to form a racket frame. Any necessary reinforcing elements an/or additional fibers and binder are added. The mold is then closed, and fluid under pressure is supplied to the interior of the casing, thereby forcing the helically wound fibers and the binder to the extremes of the mold cavity. Suitable conditions are provided so that the binder is hardened, and at a predetermined time the basic frame is removed from the mold.

It will be apparent from the foregoing discussion that the racket frame according to this invention is integrally formed by a tubular member composed of helically wound fibers of high tensile strength embedded in a hardened binder. The tubular member is curved to define an oval head portion with parallel extending portions of the tubular member defining a handle. Added reinforcements complete the oval head and strengthen the handle, which may be provided with a grip of the desired size and configuration. the more

The resulting racket possesses high strength, a good modulus of elasticity, and can be produced at low cost. Additionally, the racket of this invention can be strung and repaired by using the conventional pick and awl technique. Even when strung with a synthetic material such as nylon, the performance of the racket is comparable to that of a racket having themore expensive natural gut strings.

DESCRIPTION OF THE DRAWINGS

FIG. **1** is a perspective view of a finished racket according to this invention.

FIG. **2** is a fragmentary perspective view showing the core, the casing, and a portion of the helically wound fibers.

FIG. **3** is a perspective view of the reinforcing member.

FIG. **4** is a plan view of the mold showing additional reinforcing fibers in place.

FIG. **5** is a plan view showing the tubular member as placed in the mold.

FIG. **6** is a perspective view showing a modified racket frame.

FIG. **7** is a plan view showing diagrammatically the opposite hand, helical winding step.

FIG. **8** is a cross sectional view along the line **8—8** of FIG. **5** on a greatly enlarged scale.

DESCRIPTION OF THE PREFERRED EMBODIMENT

Referring first to FIG. **1**, a tennis racket according to this invention is shown in perspective. The frame includes a generally oval shaped head portion indicated generally at **10** and a handle portion indicated generally at **12.** The head **10** is provided with strings **14** and the handle **12** is provided with a grip **16,** both of which can be regarded as substantially conventional.

The head and handle portions are integrally formed by a tubular member **18.** The tubular member **18** is formed in a mold to provide an open oval or loop **18***a*, and the spaced apart, parallel handle portions **18***b* and **18***c* (See FIG. **5**)

The reinforcing member **20** (see separately in FIG. **3**) has the neck portion **20***a* adapted to extend across the open base of the oval defined by the tubular member **18***a*, and the handle portion **20***b* disposed between the portions **18***b* and **18***c* of the tubular member.

In the embodiment shown in FIG. **1,** the handle portion **12** is molded about the parallel extending portions **18***b* and **18***c* of the tubular member and the handle portion **20***b* of the reinforcing member. **5**).

The racket frame just described may be produced by the following process. A core seen in FIG. **2** at **30** is provided. In this Figure, the core is shown as comprising a plurality of wires. These wires may be of any material which if flexible, is capable of being placed under sufficient tension to provide a relatively rigid core for

the helical winding step to be discussed in a moment. A bundle of steel and/or aluminum wires has been proved in experimental practice to be satisfactory.

The core **30** is surrounded by a casing **32** which must be expandable, and fluid tight or impermeable. As will be apparent hereinafter, the strength requirements of the casing **32** are not great. Hence any thin and somewhat elastic material can be used. Preferably, the casing **32** will be a thin rubber material.

The casing and core just described will be placed under sufficient tension to form a relatively rigid structure, about which a plurality of layers of fiber or fiber material can be helically wound. Present practice has indicated that the most suitable material for the helical winding is a strip or ribbon of glass cloth substantially 1 inch wide shown at **34.** This material is wound back and forth over the core and casing described above in a helical pattern. This winding can be continued until the desired wall thickness is achieved. Analytical and empirical investigations have established that a wall thickness on the order of 0.050 to 0.100 inches is satisfactory from the strength and weight standpoints, and this is accomplished by helically winding six to eight layers of the glass strips.

The preferred winding pattern contemplates that the strip or ribbon be at an angle of approximately 45° to the tube axis. The fibers of one layer are thus at right angles to the fibers of the adjacent layers. This contributes to torsional stiffness and resistance to splitting.

The winding process may be facilitated by simultaneously winding two layers from opposite sides of the core and casing during each pass as shown diagrammatically in FIG. **7.** This procedure provides opposed tensions during winding making it easier to hold the core straight, and cuts winding time in half. During the winding process, additional windings may be added for reinforcement in zones of high stress.

After winding, the core **30** which is surrounded by the casing **32** and the helically wound fibers **34** is released from tension. The core **30** is preferably removed at this time, although it may be left in place until later if desired. The casing **32** and fibers **34** are then ready to be laid in a mold half such as seen in FIG. **4.**

It will be observed that the mold half has a cavity indicated generally at **40** shaped to produce the finished racket. (While only one mold half is shown, it will be understood that the two mold halves are identical.) Generally considered, the casing and helically wound fibers, the reinforcing member, any additional reinforcing fibers, and a suitable binder are placed in the mold half. The mold is then closed, fluid under pressure is introduced to the interior of the casing, so as to expand the casing and force the binder impregnated helical windings into contact with all portions of the mold cavity **40.** The binder is then cured and the racket frame is removed from the mold. Within this general framework, the precise order of steps may be varied as desired.

A desirable cross sectional configuration for the head portion of the mold cavity is seen in FIG. **8,** it again being understood that the cavities in the two mold halves are identical. The cavity shown in FIG. **8** includes the semi-cylindrical portion **41,** and the flat bottomed groove **41***a* cut out at the lowermost portion. The groove **41***a* is utilized to carry longitudinal reinforcing fibers as will be explained hereinafter. The groove **41***a* should extend around at least the entire head portion **10.**

An exemplary process contemplates that the entire mold cavity **40** is first coated with a surface coat of resin (pigmented, if desired) in order to provide an improved surface finish to the completed racket. Longitudinal reinforcing fibers **42** are laid in the groove **41***a*. Any suitable high strength fiber will be satisfactory. Excellent rackets have been made using both glass fiber strand or carbon fibers. Neck reinforcing strips **43** may then be laid on each side of each of the mold halves and coated with additional resin binder. The handle reinforcing strips **44** (which may be precut to the desired configuration) are then laid in one mold half and coated with additional binder. At this time, the casing **32** and helically wound fibers **34,** and the reinforcing member **20** are laid in position in the mold. The impregnating of the helically wound fibers **34** may conveniently be accomplished by soaking the entire strand in a resin filled trough ahead of time, or the resin may be added later. In any event, the helically wound fibers are laid into the shape described earlier, with the reinforcing member **20** in place defining the base of the oval and disposed between the portions **18***b* and **18***c* of the tubular member. The other end of the handle reinforcing strips **44** are then folded back over the reinforcing member **20** and portions **18***b* and **18***c* of the tubular member, and the mold is ready to be closed.

It will be seen in FIG. **5** that the ends of the portions **18***b* and **18***c* of the tubular member **18** extend beyond the end of the handle of the racket and to the edge of the mold. Fluid under pressure is then introduced to the interior of the casing **32.** This may be accomplished by closing off one end of the casing (for example at the free end of the portion **18***c*) and introducing air under pressure through conventional fittings to the other end. Pressures on the order of 50 to 100 psi have been found entirely satisfactory. As indicated earlier, this pressure is effective to expand the casing and force the binder impregnated helical windings into contact with all portions of the mold cavity **40.** If desired, a heated fluid can be utilized in order to expedite the hardening of a heat settable binder.

Once the binder has hardened, the mold halves are separated and the racket frame removed for further processing. The fluid tight casing **32** may be removed for reuse.

Completion of the racket frame of course requires formation of apertures for receiving the strings **14.** These apertures may be formed during the molding process by movable pins in the mold, or by drilling with conventional equipment after removal of the frame from the mold.

The further processing may desirably include the filling of the hollow tubular member with a foam core. Suitable material may be placed in the tubular member and foamed in situ according to well known techniques. This foam core helps damp vibration, and greatly facilitates stringing by guiding the end of the string across the tube.

The drawing illustrate the tubular member **18** as being substantially circular in cross section. Actual use of a racket formed in accordance with this process has been entirely satisfactory. It is of course to be pointed out that the foregoing process will permit the formation of a frame having other cross sections. That is, if it is desired to produce a racket frame having an eliptical

cross section, it is only necessary to provide the appropriate configuration for the mold cavity **40.** It will also be apparent that the mold cavity **40** may have differing cross sections, the fluid under pressure within the casing serving to cause the binder impregnated fibers to conform to whatever cavity is present.

An alternative construction is shown in FIG. **6.** According to this embodiment of the invention, the reinforcing segment **50** again extends across the open base of the oval defined by the tubular member **18***a*, but does not form a portion of the finished handle. According to this embodiment of the invention, the mold will provide a cavity to form the tubular member **18** in the general configuration described earlier; that is, an open oval or loop **18***a* and the spaced apart, parallel handle portions **18***b* and **18***c*. No provision is made for the formation of an integral handle or grip portion. Under this practice, the grip portion may be made in the form of a pair of mating grip blocks **52** and **54** which are subsequently secured to the portions **18***b* and **18***c* of the tubular member. These grip blocks may be formed of wood and the securing may be accomplished by a suitable adhesive or mechanical fasteners. It would also be possible to form these grip blocks of plastic and provide structures such that the two blocks will interlock together.

Numerous modifications can be made in the foregoing disclosure without departing from the scope and spirit of this invention. Specifically, the fiber reinforcing fiber **42** or strips **43** and **44** are exemplary only. This invention comprehends the use of additional fiber reinforcement at any desired point or section of the racket frame.

It will also be apparent that the method disclosed may be used for the production of virtually any curved, tubular article, and by providing a tapered core, the tubular member can be given a tapered configuration.

The embodiments of the invention in which an exclusive property or privilege is claimed are defined as follows:

1. A method of making a racket frame comprising the steps of:

a. providing a core which is substantially rigid in a radial direction;
b. covering said core with a flexible, fluid-tight casing;
c. placing said core under axial tension;
d. helically winding a plurality of layers of a strip of material about said core and casing to form a flexible, tubular member;
e. impregnating said material with a hardenable binder;
f. releasing said core from said axial tension;
g. placing said tubular member in a mold having a cavity defining a racket shape;
h. applying fluid under pressure to the interior of said casing;
i. removing said core from said casing after said helical winding step and before applying fluid under pressure thereto; and
j. hardening said binder.

2. The method claimed in claim **1** wherein said core is axially flexible.

3. The method claimed in claim **1** wherein said helical winding includes at least two helixes of opposite hand.

4. The method claimed in claim **1** including the steps of providing longitudinal reinforcing fibers for at least the head portion of said racket frame, and bonding said reinforcing fibers to said helically wound material.

* * * * *